TIPA Korea Technology and Information Promotion Agency for SMEs

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# Technology Proposal for 2026 TIPA* International Joint R&D Program

**(TIPA: Korea Technology and Information Promotion Agency for SMEs)**

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

## 1. Technology Proposer Information

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## 2. Technology Overview

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| Title of Proposed Technology | **Generative AI–Based CAD System for Dental Prosthetics** | | | |
| Korea's 12 National Strategic Technology Fields | Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (V ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) | | | |
| Industrial Technology Classification [1] | **Technology Field 1** (Large Category) | | **Technology Field 2** (Large Category) | |
| | *Mechanical · Materials* | | | |
| | Medium Category | Small Category | Medium Category | Small Category |
| | *Precision Manufacturing Machinery* | *CAD/CAM-related Software* | | |
| Concept of Proposed Technology | **Core Concept**<br>The proposed technology of this project is to develop a system in which a generative AI–based CAD program automatically creates dental prosthetic designs by inputting intraoral and model scanning data. Through this system, Single Crown designs can be generated in approximately 1 minute, and 3-unit bridge designs in approximately 5–7 minutes, enabling a production structure capable of completing 20 designs per day within one hour.<br>Through this generative AI CAD system, the project aims to minimize variations in productivity and quality caused by individual skill levels, and to free existing personnel from repetitive design tasks, thereby achieving overall productivity improvement through effective workforce reallocation.<br><br>**Key Functions and Potential Applications**<br>1) Digitalization of Intraoral Scan Data and AI Input Module | | | |

- 3D scan data (e.g., STL, OBJ) acquired from a patient's oral cavity or dental models are used as the initial input to the system.
- Beyond simple 3D geometry, the AI performs detailed analysis of the intraoral environment—such as interproximal spacing, occlusal relationships, and adjacent tooth morphology—to construct patient-specific data from the earliest design stage.

2) Generative AI Core Engine
   - A deep learning–based generative AI model.
   - Advanced AI techniques such as Generative Adversarial Networks (GANs) or Diffusion Models are employed to enable high-level design generation.
   - The model is trained on high-quality clinical design data previously created within individual dental laboratories.
   - Through training, the AI creatively generates optimized designs that reflect the unique "character" of each dental laboratory.
   - The concept involves digitizing and embedding the accumulated know-how and expertise of skilled dental technicians, built over decades, into the AI model.

3) Ultra-Fast Automatic Design Generation Process
   - When scan data are input into the AI core engine, patient-specific prosthetic design drafts (e.g., single crowns, bridges) are automatically generated.
   - Design tasks that previously required several minutes to tens of minutes using manual methods are dramatically reduced to tens of seconds to within one minute, achieving a revolutionary improvement in design productivity.

4) Optimization and Precision Validation Algorithms
   - Automatic optimization and precise validation of the generated designs.
   - The system independently simulates and evaluates occlusal accuracy, esthetic factors, and functional suitability in an integrated manner.
   - Potential occlusal interferences are identified in advance, and the system proposes the most ideal design by considering optimal reduction amounts, prosthesis thickness, and margin fitting.
   - This minimizes the possibility of human error and maximizes final design quality.

5) Integration with Existing CAD Software and Scalability
   - The AI design system can be provided as a standalone solution or integrated as a plug-in with major CAD software widely used in the dental laboratory market, such as exocad and 3Shape.
   - This flexibility reduces the learning burden for users and enables immediate use of AI-powered design functions within existing workflows, supporting seamless and efficient operations.

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| **Rationale for Joint R&D Project** | **Critical Technical Barriers**<br>In the current dental prosthesis manufacturing process, intraoral and model scanning data are manually designed by dental technicians using CAD software, requiring an average of 7–10 minutes for a single crown and 20–30 minutes for a 3-unit bridge. As a result, each technician typically spends 4–5 hours per day to complete approximately 20 designs, with significant variations in productivity and output quality depending on individual skill levels. In addition, the ongoing decline in the dental technician workforce has further intensified difficulties in securing sufficient production personnel.<br>At the same time, AI and medical device technologies are advancing rapidly worldwide, making it increasingly difficult for any single country or institution to independently secure and maintain all cutting-edge technologies. Moreover, the performance of AI models is highly dependent on the volume, quality, and diversity of training data, highlighting the importance of large-scale and diverse datasets for reliable automation and standardization of dental design processes. Furthermore, dental medical devices are subject to different regulatory and approval requirements across countries, underscoring the need to consider global regulatory environments alongside technological innovation.<br><br>**Why Global Collaboration?**<br>1) Securing Global Leading Technologies and Strengthening Competitive Advantage<br>● Through collaboration with leading overseas research institutions and AI-specialized companies, the project aims to rapidly acquire and integrate state-of-the-art AI algorithms, large-scale clinical data processing technologies, and region-specific design know-how, thereby securing a competitive advantage and establishing global technological leadership.<br>● Close collaboration with, and effective utilization of, outstanding international talent is a key strategic focus.<br>2) Acquisition of Diverse and Large-Scale Datasets and Advancement of AI Models<br>● By securing international-scale datasets that reflect country-specific oral anatomies, ethnic characteristics, and dental treatment trends, the project can maximize the generalization performance of AI models and develop highly versatile design solutions that are not limited to specific regions.<br><br>**Expected Impact**<br>1) Acquisition of Diverse and Large-Scale Datasets and Advancement of AI Models<br>● This approach is expected to serve as a foundation for a significant improvement in the precision and robustness of AI models.<br>2) Proactive Preparation for Global Standards and Regulatory Compliance<br>● Through international collaborative R&D, the project will work with regulatory experts from the development stage, validate clinical effectiveness in real-world environments, and establish strategies for obtaining required certifications (e.g., CE, FDA), thereby minimizing uncertainties in global market entry and facilitating successful market adoption. |

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| **Key Words** | 1. AI Customization<br>2. AI Personalization<br>3. High Productivity<br>4. Workforce Reallocation<br>5. Workflow Integration |

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## 3. Research and Development Objectives & Plan

| Technology Readiness Level (TRL)[2] | Start of the Project | Level ( 3 ) | End of the Project | Level ( 7 ) |
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| R&D Objectives | **R&D Objectives**<br>1) Development and Performance Optimization of the AI Core Design Engine<br>● To develop a generative AI model (based on GANs/Diffusion Models) that automatically generates optimal designs for single crowns and 3-unit bridges or larger, by inputting patient intraoral scan data and considering anatomical characteristics, occlusal relationships, and esthetic requirements.<br>2) Design Automation and Drastic Reduction of Design Time<br>● To significantly reduce manual dental prosthesis design tasks and maximize automation, achieving a dramatic reduction in design time.<br>3) Design Quality Standardization and Minimization of Error Rates<br>● To eliminate variations in design quality caused by individual skill levels and ensure that all generated designs meet clinical standards and esthetic requirements, achieving standardized high quality.<br>4) High Compatibility with Existing CAD Software and Development of a User-Friendly Interface<br>● To ensure high compatibility with widely used commercial CAD software in dental laboratories and provide an intuitive user interface (UI) and user experience (UX) that enables easy utilization of AI design functions.<br><br>**Detailed Objectives**<br>1) Development and Performance Optimization of the AI Core Design Engine<br>● Construction of a large-scale, high-quality training dataset incorporating domestic and international clinical data as well as the design know-how of experienced dental technicians, and enhancement of AI model precision and diversity.<br>● Performance optimization to ensure that the developed AI model can generate clinically appropriate designs across a wide range of cases.<br>2) Design Automation and Drastic Reduction of Design Time<br>● Optimization of AI processing speed to complete single crown design drafts within 1 minute and 3-unit bridge design drafts within 5 minutes.<br>● Increasing overall workflow automation by enabling the AI to lead the entire process, including design planning, draft generation, and initial review.<br>3) Design Quality Standardization and Minimization of Error Rates<br>● Development of functions that automatically detect major design errors—such as occlusal interference and margin misfit—and provide correction guidelines or perform automatic corrections. | | | |

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| **R&D Target Specifications** | ● System design to ensure that prostheses generated by the platform consistently achieve high levels of fit and functional performance.<br>4) High Compatibility with Existing CAD Software and Development of a User-Friendly Interface<br>● Development of plug-in integration modules for leading CAD software with high market share, such as exocad and 3Shape.<br>● Design and implementation of intuitive UI/UX that allow easy review, modification, and final approval of AI-generated designs. |
| | 1) Development and Performance Optimization of the AI Core Design Engine<br>● Achieving clinical suitability of AI-generated designs of at least 95%, based on expert evaluation.<br>● Achieving a design success rate of at least 90% across diverse cases.<br>2) Design Automation and Drastic Reduction of Design Time<br>● Achieving a reduction of design time of at least 90% compared to conventional manual design processes.<br>● Achieving automation of at least 90% of the overall design workflow.<br>3) Design Quality Standardization and Minimization of Error Rates<br>● Maintaining a design error rate below 5% (with an initial target of within 10%).<br>● Maintaining a design rework rate below 3%.<br>4) High Compatibility with Existing CAD Software and Development of a User-Friendly Interface<br>● Achieving an integration rate of at least 90% with major commercial CAD software platforms.<br>● Achieving user satisfaction of at least 85%, based on usability and convenience evaluations. |
| **R&D Plan** | **Year 1: Completion of AI core engine development and establishment of the system foundation**<br>● Securing high-quality training datasets<br>● Development of the AI core model for single crowns<br>● Development of basic CAD integration modules<br>● Construction of an internal prototype for testing<br>● Internal performance verification<br><br>**Year 2: Functional expansion, performance enhancement, and development of a user-friendly system**<br>● Expansion of the AI engine to support multi-unit designs (e.g., bridges)<br>● Enhancement of automatic design optimization functions<br>● Development of design error detection and correction functions<br><br>**Year 3: Final system completion** |

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| | ● Design and implementation of user UI/UX<br>● Pilot testing with external organizations<br>● Final system verification and stabilization |

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## 4. R&R (Roles and Responsibilities)

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| **Expected R&R of Research Institute** | 1) Development of the AI core module and program<br>● Development of the AI core design module and the overall program using state-of-the-art deep learning–based generative AI algorithms (e.g., GANs, Diffusion Models).<br>2) AI model training and performance enhancement<br>● Training of AI models based on data provided by domestic companies, along with performance enhancement and algorithm improvements to achieve clinical and technical objectives.<br>3) Technical design and architecture development<br>● Design and implementation of the overall technical architecture of the AI design system, ensuring system stability, scalability, and efficiency. |
| **Expected R&R of Korean SME Partner** | 1) Data acquisition and management<br>● Acquisition and processing of high-quality dental prosthetic design data required for AI model training (e.g., intraoral scans and finalized designs), refinement into AI-ready formats, and provision of the processed data.<br>2) Market requirements analysis<br>● Analysis of functional requirements and features of the AI design system based on the needs of the domestic dental laboratory market and feedback from practicing dental technicians and dentists, supporting product planning and decision-making.<br>3) Clinical validation and feedback<br>● Clinical validation of the developed AI design system and generated outputs (e.g., accuracy and fit), and provision of regular feedback to overseas university research teams.<br>4) Localization and user support<br>● Development of localization strategies for the domestic dental environment, establishment of technical support systems for users, and development of user training programs.<br>5) Contribution to intellectual property acquisition<br>● Contribution to the acquisition and management of intellectual property through joint R&D, and development of technology protection strategies. |

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## 5. Expected Outcomes/Impact

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| **Expected Outcomes and Impact** | **Direct Expected Outcomes**<br>1) Dramatic productivity improvement<br>● Achieving design times of within 1 minute for single crowns and within 5 minutes for 3-unit bridges, resulting in an overall increase in design productivity of more than 300% for dental laboratories.<br>2) Standardization of design quality<br>● Providing consistently high-quality designs based on AI, completely eliminating quality variations caused by individual skill levels and achieving a design error rate of less than 5%.<br>3) Cost reduction effects<br>● Reduction of labor and material costs through shortened design time and decreased rework rates, leading to improved profitability for dental laboratories.<br>4) Securing technological patents and intellectual property<br>● Filing domestic and international patents and strengthening intellectual property protection for core technologies, including AI model training and algorithm optimization, as well as UI/UX.<br>5) Enhancement of dental laboratory competitiveness<br>● Strengthening market competitiveness by increasing dental prosthesis production volume and delivering high-quality products through the adoption of AI systems.<br>6) Strengthening professional workforce capabilities<br>● Reducing the burden of repetitive tasks for dental technicians and enabling them to focus on high value-added design work leveraging AI.<br><br>**Industrial and Social Impact**<br>1) Acceleration of digital transformation in the dental laboratory industry:<br>● Accelerating digitalization and the transition to smart manufacturing processes across the dental laboratory industry through the adoption of AI-based design solutions.<br>2) Alleviation of workforce shortages among dental technicians:<br>● Addressing labor shortages through productivity improvements, improving working conditions to prevent workforce attrition, and encouraging new workforce entry.<br>3) Improved patient satisfaction and quality of care:<br>● Enhancing patient treatment experiences and satisfaction by providing faster, more accurate, and more esthetically superior dental prostheses.<br>4) Strengthening global competitiveness:<br>● Enhancing the international standing of domestic digital dentistry technologies and securing a competitive advantage in global markets through the development of innovative AI-based solutions. |

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| | 5) Presentation of an advanced technology convergence model:<br>● Demonstrating a successful convergence of dental healthcare and AI technologies, serving as a model for technology integration across other medical fields.<br>6) Creation of new employment opportunities:<br>● Creating new specialized jobs in related fields, including AI system operation and maintenance, data management, and AI-CAD training. |

## 6. References

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